

Experiment: Downloading, Installing, and Configuring OpenJDK & Oracle JDK SE 26 on Ubuntu

Aim

To download, install, and configure OpenJDK and Oracle JDK SE 26 on the Ubuntu operating system and verify the installation.

Requirements

- Ubuntu Operating System
- Internet connection
- Terminal access
- Oracle JDK SE 26 installation package
- Administrator (sudo) privileges

Procedure

Part A: Installing OpenJDK

Step 1: Update the package list.

```
sudo apt update
```

Step 2: Install OpenJDK.

```
sudo apt install openjdk-21-jdk -y
```

(If your Ubuntu repository provides a different version, install that version instead.)

Step 3: Verify the installation.

```
java -version
```

Step 4: Check the Java compiler version.

```
javac -version
```

Part B: Downloading Oracle JDK SE 26

Step 1: Open a web browser.

Step 2: Go to the official Oracle Java Downloads page.

Step 3: Download the Linux x64 Archive (.tar.gz) for Oracle JDK SE 26.

Step 4: Save the downloaded file in the Downloads folder.

Part C: Installing Oracle JDK SE 26

Step 1: Open the terminal and move to the Downloads folder.

```
cd ~/Downloads
```

Step 2: Create the installation directory.

```
sudo mkdir -p /usr/lib/jvm
```

Step 3: Extract the downloaded archive.

```
sudo tar -xvzf jdk-26_linux-x64_bin.tar.gz -C  
/usr/lib/jvm
```

Part D: Configuring Oracle JDK

Step 1: Open the system profile.

```
sudo nano /etc/profile
```

Step 2: Add the following lines at the end of the file.

```
export JAVA_HOME=/usr/lib/jvm/jdk-26  
export PATH=$JAVA_HOME/bin:$PATH
```

Save the file and exit.

Step 3: Reload the profile.

```
source /etc/profile
```

Part E: Set Oracle JDK as the Default Java

Run:

```
sudo update-alternatives --config java
```

Select the Oracle JDK option.

Similarly, configure the Java compiler.

```
sudo update-alternatives --config javac
```

Part F: Verify the Installation

Check the installed Java version.

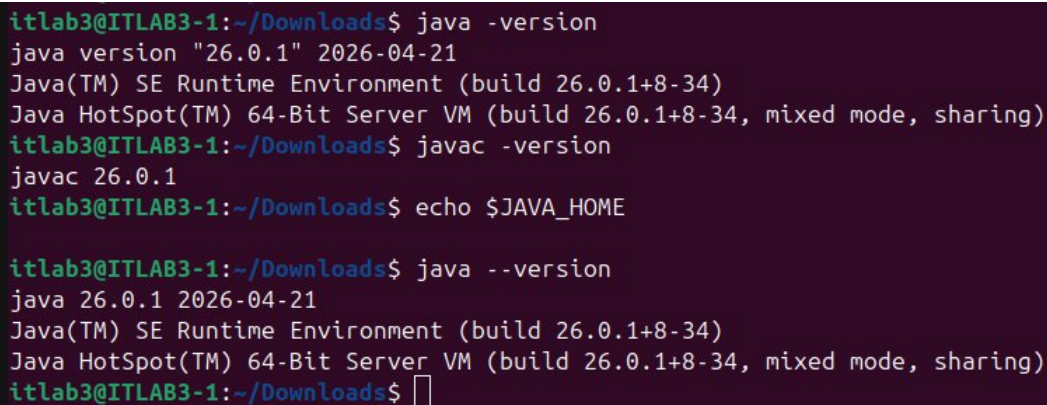
```
java -version
```

Check the compiler version.

```
javac -version
```

If the version displayed is Oracle JDK SE 26, the installation is successful.

Screenshot

A terminal window with a dark purple background and light green text. The prompt is 'itlab3@ITLAB3-1:~/Downloads\$'. The first command is 'java -version', which outputs: 'java version "26.0.1" 2026-04-21', 'Java(TM) SE Runtime Environment (build 26.0.1+8-34)', and 'Java HotSpot(TM) 64-Bit Server VM (build 26.0.1+8-34, mixed mode, sharing)'. The second command is 'javac -version', which outputs: 'javac 26.0.1'. The third command is 'echo \$JAVA_HOME', which outputs an empty line. The fourth command is 'java --version', which outputs the same information as the first command. The prompt is now 'itlab3@ITLAB3-1:~/Downloads\$' with a cursor.

```
itlab3@ITLAB3-1:~/Downloads$ java -version
java version "26.0.1" 2026-04-21
Java(TM) SE Runtime Environment (build 26.0.1+8-34)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.1+8-34, mixed mode, sharing)
itlab3@ITLAB3-1:~/Downloads$ javac -version
javac 26.0.1
itlab3@ITLAB3-1:~/Downloads$ echo $JAVA_HOME

itlab3@ITLAB3-1:~/Downloads$ java --version
java 26.0.1 2026-04-21
Java(TM) SE Runtime Environment (build 26.0.1+8-34)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.1+8-34, mixed mode, sharing)
itlab3@ITLAB3-1:~/Downloads$
```

Result

OpenJDK and Oracle JDK SE 26 were successfully downloaded, installed, configured, and verified on the Ubuntu operating system. The Java environment was successfully configured and the installed version was confirmed using the `java -version` and `javac -version` commands.